PROJECT REPORT

SECJ2154

SECTION 02

OBJECT ORIENTED PROGRAMMING

MINI PROJECT TITLE:

STAFF ATTENDANCE APPLICATION

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1.0 INTRODUCTION

The **Staff Attendance Application** is a Java-based system designed to automate the attendance tracking process for organizations. It replaces traditional manual methods, such as paper logbooks or spreadsheets, which are often inefficient and prone to error. By applying object-oriented programming (OOP) principles, such as inheritance, abstraction, polymorphism, and interfaces, the application is structured for clarity and scalability.

The system allows staff to clock in using a menu-driven console interface. It records the exact date and time of attendance using timestamps and stores the information in text files through file handling techniques. This ensures that attendance records are both accurate and permanently saved. The system also distinguishes between full-time and part-time staff using separate classes, making it more adaptable to real workplace settings. Therefore, the system improves accuracy, reduces human error, ensures data is permanently saved, and offers a more organized and adaptable approach to managing staff attendance.

2.0 PROBLEM STATEMENT

Many organizations still rely on manual methods to record staff attendance, such as physical logbooks or basic spreadsheets. These methods are often inefficient, lack proper input validation, and are vulnerable to data manipulation or loss. Additionally, they do not provide real-time tracking, make it difficult to differentiate between full-time and part-time staff, and offer limited options for generating accurate reports.

To overcome these limitations, the Staff Attendance Application was developed. It offers an automated solution that:

- Differentiates between full-time and part-time staff attendance
- Uses Timestamp to accurately log clock-in times
- Implements file handling for secure and persistent data storage
- Validates input using exception handling to ensure data accuracy

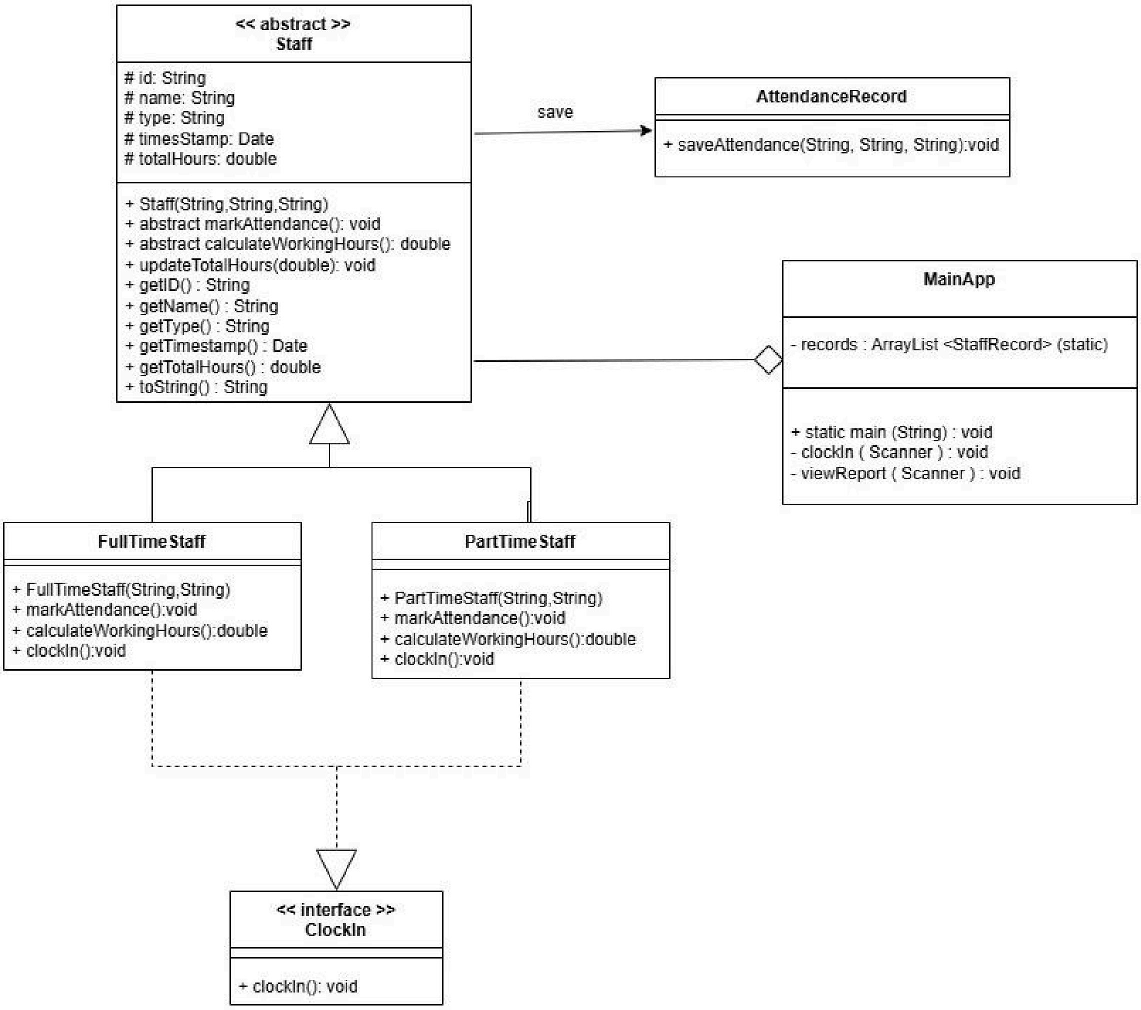
This system addresses the shortcomings of manual processes by providing a reliable and structured approach to staff attendance tracking.

3.0 OBJECTIVE

The main objectives of this project are:

- To develop a staff attendance system that automates clock-in processes using Java.
- To apply object-oriented programming concepts such as inheritance, abstraction, polymorphism, and interfaces for better code structure and maintainability.
- To differentiate between full-time and part-time staff through subclassing and behavior customization.
- To accurately record attendance using Timestamp for real-time date and time logging.
- To implement file handling for persistent and secure storage of attendance records.
- To ensure reliable user input through exception handling and validation techniques.
- To provide a simple, menu-driven console interface for ease of use and interaction.

4.0 PROJECT DESIGN (CLASS DIAGRAM)



5.0 PROJECT OUTPUT (SCREENSHOTS)

1. Menu Staff Attendance System

```
=== Staff Attendance System ===

Menu:
1. Clock In
2. View Attendance Report
3. Exit
Choose an option (1-3):
```

2. Clock In for Full-Time Staff

```
Choose an option (1-3): 1
Enter Staff ID: 001
Enter Staff Name: naim
Is this Full-Time staff? (yes/no): yes
Enter OT hours (0 if none): 2
Full-time staff naim clocked in.
Marking full-time attendance for: naim
Attendance saved.

Working Hours (including OT): 10.0
```

3. Clock In for Part-Time Staff

```
Choose an option (1-3): 1
Enter Staff ID: 002
Enter Staff Name: ayam
Is this Full-Time staff? (yes/no): no
Enter OT hours (0 if none): 5
Part-time staff ayam clocked in.
Marking part-time attendance for: ayam
Attendance saved.

Working Hours (including OT): 9.0
```

4. View Attendance Report (Full-Time)

```
Choose an option (1-3): 2
View which type of staff? (full/part): full

--- FullTime Attendance Report for 2025-06-25 ---
ID          Name              Type          Date        Time        Hours
001         naim              FullTime      2025-06-25 20:23:32 10.00
```

5. View Attendance Report (Part-Time)

```
View which type of staff? (full/part): part

--- PartTime Attendance Report for 2025-06-25 ---
ID          Name              Type          Date        Time        Hours
002         ayam              PartTime      2025-06-25 20:24:24 9.00
```

6. Exit the Staff Attendance Report

```
Choose an option (1-3): 3
System exited.
```

6.0 CONCLUSION

In conclusion, by completing this staff attendance application system project, the manual method of recording staff attendance has been replaced with a Java code, which is resulting in more efficiency, improved validation and protection against data manipulation or loss. With this system, the admin could monitor staff attendance based on their employment type such as full-time or part-time. Additionally, by using timestamps, we could prevent the employee from faking their clock-in timings. This is because the administrator will be able to identify it if the recorded clock-in time differs from the specified working hours. Moreover, if the admin chooses not to access the system directly, he/she can still view the attendance record by checking it in the notepad file. We implemented the file handling for more secure and persistent data storage. Not only that, we also have implemented exception handling to ensure that all data that we received is accurate. Lastly, all these functions work really well because we have applied the object-oriented programming concepts such as inheritance, polymorphism, and interface in order to develop our system that is more efficient and systematic in the attendance tracking system.

7.0 APPENDICES

Link code files:
https://drive.google.com/file/d/14gRXyaX1ZJokHMNXgt8t1Ld_n86rBtDf/view?usp=sharing